

Exam date: April 14

1. Interconnected state-space models to overall state-space model.
2. Observers
 - (a) Observer state-space equations and observer settling time.
 - (b) Calculation of observer gain vector using `place` for single-output systems to achieve a desired observer settling time (e.g. a 4-times faster observer).
 - (c) Observer-based regulators
3. Characterization of solutions to homogeneous vector difference equations, $\mathbf{x}[k+1] = \mathbf{A}\mathbf{x}[k]$, $\mathbf{x}[0] = \mathbf{x}_0$.
4. Understanding what the `place` function does.
5. Finding a state-space model for the loop gain of a given feedback control system. (This model is given to the `dsm` function to calculate stability margins.)
6. The meaning of stability margins; in particular, gain margin. How gain margin relates to perturbations of the plant model for which stability of the closed-loop system is guaranteed.
7. Understanding what an observer does.
8. The block diagram for an observer-based regulator.

1. A control system that is stable with a nominal plant model $(\mathbf{A}, \mathbf{B}, \mathbf{C})$ will remain stable with a perturbed plant model $(\mathbf{A}, q\mathbf{B}, \mathbf{C})$ if q is a real number such that

$$q_{min} < q < q_{max}$$

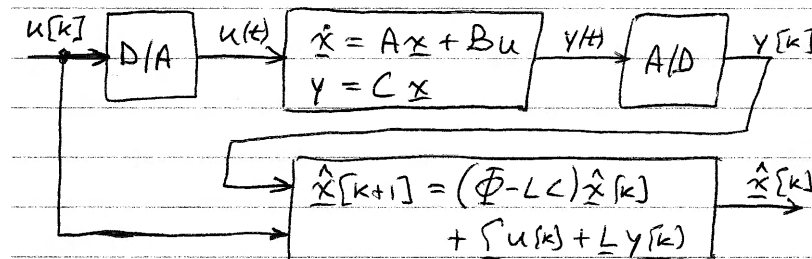
or if

$$q = e^{j\phi} \text{ and } |\phi| < \phi_{max}.$$

The lower gain margin is $20 \log_{10}(q_{min})$ dB. The upper gain margin is $20 \log_{10}(q_{max})$ dB. The phase margin is ϕ_{max} . The function that converts a number x to dB is inverted by the following formula

$$x = 10^{\frac{\text{dB}}{20}}.$$

2. Classical stability margins for a digital control system may be found by giving a state-space model of the loop system to the Matlab function `dsm`. The loop system is found using the following steps: set all external inputs to zero; break the loop open at the plant input $\mathbf{u}$; insert a minus sign on the loop output and call the resulting signal $\mathbf{w}$; find a state-space model from $\mathbf{u}$ to $\mathbf{w}$.
3. `K=place(A,B,e)` calculates a matrix $\mathbf{K}$ such that the eigenvalues of $\mathbf{A} - \mathbf{BK}$ are the numbers specified in the vector $\mathbf{e}$.
4. A (digital) observer takes discrete-time input and output sequences from a given plant and produces estimates of the plant state vector. The observer estimates converge to samples of the plant state vector if all of the eigenvalues of $\Phi - \mathbf{LC}$ are inside the unit circle.



5. An observer-based regulator uses an observer to produce $\hat{\mathbf{x}}[k]$, an estimate of the plant state vector $\mathbf{x}[k]$. The estimated plant state vector is fed back to the plant input through the feedback gain vector $\mathbf{K}$. Because the plant input is now a function of the observer state vector, $u[k] = -\mathbf{K}\hat{\mathbf{x}}[k]$, $u[k]$ no longer has to be sent to the observer as an input. Rather, it is absorbed into the state update equation, which becomes $\hat{\mathbf{x}}[k+1] = (\Phi - \mathbf{LC} - \Gamma\mathbf{K})\hat{\mathbf{x}}[k] + \mathbf{L}y[k]$.
6. The observer settling time, T_{so} , is chosen to be “three to five times faster” than T_s , the desired settling time for the regulator. That is: $T_s/5 \leq T_{so} \leq T_s/3$. The observer poles are chosen using scaled Bessel poles, `sopoles=sn/Tso`, where n is the order of the plant model. For digital observers, the analog observer poles must be mapped into digital observer poles, `zopoles=exp(T*sopoles)`.